

C++ Post Training Final Assessment

Test Reference No: PLMAcademy-Nov2025-General-Batch

Total Marks : 30

Test Duration : 1 hour

Instructions to Candidate

1. All questions are compulsory/mandatory.
2. Do not write anything on the question paper.
Write everything on the answer sheet provided.
3. This is a closed book test.
4. There is no negative marking.
5. Wherever there is ambiguity, make suitable assumptions and proceed.
Document the assumptions in the answer.
6. Assume C++ 11 language standard.

Assume usage of Microsoft Visual Studio development platform.

Question No.	Question	Marks
1	What will be the output of the following: <pre>#include <iostream> using namespace std; class CEmployee{ public: virtual void raiseSal() { cout << "Employee salary" << endl; } virtual ~CEmployee() { cout << " In CEmployee Destructor" << endl; } }; class CManager : public CEmployee{ public: void raiseSal() { cout << "Manager salary" << endl; } ~CManager(){ cout << " In CManager Destructor" << endl; } }; int main() { CEmployee* p = new CManager(); delete p; }</pre>	1
2	Explain copy constructor and assignment operator using minimal code.	2

3	Can the constructor be made virtual in C++? What will be the output of the following code? <pre>#include <iostream> using namespace std; class X { public: virtual X() { cout << "X constructor called" << endl; } }; int main () { X x1; return 0; }</pre>	2
4	Write code so that the output in the following code is: <pre>20 30</pre> You have to write the function signature for f. No change to be made to function main. <pre>void f(_____) { } int main() { int i = 20; cout << i << endl; f(i); cout << i << endl; return 0; }</pre>	1
5	What is the output of the following program? <pre>#include <iostream> using namespace std; int main() { int a[] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}; cout << sizeof(a) / sizeof(int) << endl; return 0; }</pre>	1

6	Show the vtable entries, which are pointed to by the vptrs in objects of class X, and class Y, in the following example: <pre> class X { public : virtual int f () ; virtual int g () ; }; class Y { public : int g () ; }; </pre>	2
7	What will be the output of the following program? <pre> #include <iostream> using namespace std ; int main () { int i = 10 ; int *pi = &i ; *pi = 100 ; cout << i << endl ; return 0 ; } </pre>	1
8	What is the output of the following code? Explain the reason. <pre> #include <iostream> using namespace std ; int main() { int i = 10 ; int j = 20 ; int *pi = &i ; int *pj = &j ; if(pi == pj) { cout << "Address of pi and pj are same" << endl ; } else { cout << "Address of pi and pj are different" << endl ; } return 0 ; } </pre>	1

9	Write a small piece of code which demonstrates the usage of a static integer variable.	1
10	What is difference between unique pointer and shared pointer?	2
11	You are given a file client.cpp which contains the following code: <pre>int main() { }</pre> File client.cpp What command you will run, to generate an object file client.obj . Once the client.obj is generated, what command you will run to generate client.exe .	2
12	Will the following code correctly give the length of the linked list? If Not, correct the code. <pre>int length(Node* head) { int cnt = 0; while (head->next != NULL) { cnt++; head = head->next; } return cnt; }</pre>	1
13	Write a program to merge 2 character array alternatively.	2
14	What is Singleton Design Pattern. Write the code for the same.	2
15	Explain Dynamic casting with minimal code.	2
16	Write a program to find the second largest element in a vector.	2
17	What is a recursive function? Explain with minimal example.	1
18	What is the output of the following code: <pre>#include <iostream> using namespace std; class CEmployee{ public: virtual void raiseSal() { cout << "Employee salary" << endl; } }; class CManager : public CEmployee{</pre>	1

	<pre> public: void raiseSal() { cout << "Manager salary" << endl;}; int main() { CEmployee* p = new CManager(); p->raiseSal();} </pre>	
19	What will be the output of the following code? <pre> void f(int* p) { std::cout << "non-const\n"; } void f(const int* p) { std::cout << "const\n"; } int main() { int x = 10; const int y = 20; f(&y); f(&x); } </pre>	1
20	Mention SOLID principles with one line explanation for each.	2